

# Zi En Tham (Zen)

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## Education

**Bachelor of Computing Science (Honours)**

**August 2021 - August 2024**

**University of Technology Sydney (UTS)**

- Major: Artificial Intelligence and Data Analytics (WAM 86.58)

## Skills

**Programming language:** SQL, Python, HTML/CSS, Java, PHP, Javascript, C++

**Languages:** English, Chinese (Native), Cantonese, Malay

**Tools:** Git, Node, React, TypeScript, Docker, Laravel, Excel, Tableau, PowerBI

## Work Experience

**Web Developer**

**May 2025 – Present**

**Trivesta, Sydney, NSW, Australia**

- **Docker Compose containerization** with Traefik edge routing, Nginx reverse proxy, MariaDB, and Redis object caching
- **GitHub Actions CI/CD** reducing deployment effort by 50%
- **Docker image optimization** improving WordPress performance by 75% and cutting infrastructure costs by 30% through lower-tier VPS migration
- **Cloudflare security hardening** and DNS management
- **AI/ML predictive modeling** achieving 80%+ churn prediction accuracy to inform retention strategies
- **WordPress custom theme development** (PHP, MySQL) for corporate website and CRM platform
- **React/TypeScript frontend architecture** (Vite, Redux, Axios, Tailwind CSS) for scalable full-stack delivery

**Operation Developer**

**September 2024 - Present**

**Flash Media Solutions, Rydalmere, NSW, Australia**

- **Client relationship management** as primary point of contact for inbound inquiries and call handling; oversee inventory control and delivery logistics coordination
- **Custom WordPress theme development** (PHP, MySQL) for corporate website; reduced deployment effort by 30% through Git version control implementation
- **React-based CRM architecture** for inventory management, automated warranty generation, and delivery cost comparison modelling
- **Legacy system modernization** migrating Magento 1.0 to custom WordPress theme (PHP, MySQL) and transitioning Linux enterprise servers to new provider
- **Server infrastructure management** with automated backup pipelines ensuring business continuity

**Technical Support Engineer Intern**

**January 2020 – Mar 2020**

**Net Onboard Sdn Bhd, KL, Malaysia**

- **Network cabling infrastructure:** created and organized structured cabling for server deployments
- **Server provisioning and deployment:** prepared and deployed bare-metal and virtual servers per client requirements
- **Linux server troubleshooting:** diagnosed and resolved server issues reported by clients across Ubuntu and CentOS environments
- **Testing environment deployment:** built isolated staging environments for pre-production validation
- **Network architecture design:** planned and implemented network topologies and infrastructure layouts
- **Technical documentation:** recorded meeting minutes and maintained documentation for technical support department
- **PowerShell scripting and automation** for system administration and deployment tasks

## Projects

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### 2D Drawing to 3D Model Web Tool

Ongoing 2026

- Built browser-based tool converting 2D canvas drawings into exportable 3D GLB models
- Implemented Three.js extrusion pipeline with direct compatibility for major game engines
- Designed for mobile and desktop with future skeletal rigging support planned (ongoing)
- Tools: Three.js, JavaScript, WebGL, GLB Export

### AI Study Assistant

Ongoing 2026

- Built LLM-powered study platform with PDF upload and personalized concept explanation
- Implemented automated flashcard generation with accurate citation tracking per source
- Centralized fragmented study workflow previously spread across multiple AI chat platforms
- Tools: React, TypeScript, NVIDIA NIM API, PDF Processing, LangChain

### Gamify Physiotherapy (UNSW Hackathon 2025)

November 2025

- Built a web app helping patients track and monitor knee rehabilitation exercises
- Integrated MediaPipe pose analysis for real-time exercise form validation feedback
- Developed full-stack solution within a hackathon timeframe at UNSW Kensington
- Tools: Next.js, TypeScript, TailwindCSS, Shadcn UI, Prisma ORM, PostgreSQL, Python, MediaPipe

### Online Leave Management System

July 2023

- Built full-stack Laravel application replacing a manual student leave request process
- Implemented role-based access control for student applicants and lecturer approvers
- Automated approval workflow reducing leave processing to a same-session turnaround
- Tools: Laravel, PHP, MySQL, Bootstrap, Blade

### Inventory Management System

October 2025

- Designed normalized Excel VBA database to replace a broken legacy CSV system
- Automated stock validation, warranty tracking, and aging inventory alert generation
- Eliminated duplicate records across 30 product SKUs entirely
- Tools: Excel VBA, Database Design, CSV Processing

## Referees

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Dr Xu Wang

Senior Lecturer at School of Electrical and Data Engineering UTS

[Contact details available on request]

Eugene Nguyen

Formal Colleague at Trivesta

[Contact details available on request]